

MUSAWANDILE SHAMASE

ELECTRONICS & COMPUTER ENGINEERING GRADUATE (BENGTECH)

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PROFESSIONAL SUMMARY

Driven Electronics and Computer Engineering (BEngTech) graduate from Durban University of Technology with a strong technical foundation in embedded systems, electronic circuit design, control systems, and software development. Hands-on experience in designing and prototyping engineering projects using microcontrollers, sensors, and low-level embedded programming. Currently advancing specialized industry competencies through a PLC Programming Certification and the FNB App Academy. Seeking to leverage engineering problem-solving, hardware design, and programming expertise in a junior engineering, industrial automation, or graduate development role.

EDUCATION

Bachelor of Engineering Technology (BEngTech) in Electronics & Computer Engineering **Graduated: 2025**
Durban University of Technology (DUT), Durban, South Africa

TECHNICAL SKILLS

Programming Languages:	Python, C/C++, Arduino C++, HTML & CSS, Ladder Logic (In Progress)
Engineering & Hardware:	Embedded Systems, Circuit Design & Troubleshooting, PCB Assembly (SMT & Through-Hole), Soldering, Digital & Analog Electronics, Control Systems
Industrial Automation:	PLC Architecture, Scan Cycles, Digital & Analog I/O Configuration, Ladder Logic, Timers & Counters, Automation Fundamentals
Software & Tools:	MATLAB, Arduino IDE, Linux, Windows, macOS, Microsoft Office Suite
Instruments & Testing:	Oscilloscope, Digital Multimeter, Calibration & Testing Equipment

ENGINEERING PROJECTS

Smart Home Irrigation System	Arduino, C++, Soil Sensors, Relay Modules
- Designed and prototyped an automated irrigation control system that monitors soil moisture levels in real time to regulate water pump actuation. - Programmed microcontrollers in C++ to optimize irrigation intervals, minimizing resource waste and improving system efficiency.	
Cable Fault Detection System	Embedded Systems, Signal Processing, Sensors
- Engineered a diagnostic system capable of detecting and pinpointing fault locations in electrical and signal cabling to reduce operational downtime. - Integrated embedded sensor interfaces and signal processing logic to deliver precise fault detection and reporting.	
Car Safety Interlock System	Microcontrollers, Embedded C, Safety Logic
- Developed an embedded safety interlock system designed to inhibit vehicle startup until safety checks (such as seatbelt fastening) are validated. - Configured microcontroller hardware interrupts and firmware logic in Embedded C for real-time safety condition verification.	

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

PLC Programming Certification	In Progress
Key Focus Areas: PLC Architecture, Scan Cycles, Digital/Analog I/O, Ladder Logic, Timers & Counters, Industrial Automation	
FNB App Academy	In Progress
Key Focus Areas: Software Development Fundamentals, Programming Concepts, Application Development, Software Engineering Principles	

WORK EXPERIENCE

Merchandiser	April 2026 – Present
MyChina, Boksburg, South Africa	
- Maintain product displays, stock replenishment, and inventory accuracy in a high-volume retail environment. - Demonstrate operational discipline, attention to detail, and effective team coordination to consistently achieve daily targets.	

Educational Assistant**June 2025 – November 2025***KwaBulawayo High School (Department of Basic Education), South Africa*

- Supported educators with lesson preparation, classroom administration, and academic tracking.
- Delivered structured academic guidance and tutoring to high school learners in core STEM subjects.

Volunteer Programming Tutor**March 2022 – August 2024***Dataholics Student Organisation, Durban, South Africa*

- Tutored undergraduate students in computer programming fundamentals, logic design, and software debugging.
- Conducted practical coding workshops and test preparation sessions to enhance academic outcomes.

ADDITIONAL INFORMATION

Driver's Licence: Code 10 (C1) Driver's Licence
Relocation: Open and available to relocate across South Africa
Work Eligibility: South African Citizen (Eligible to work immediately)

REFERENCES

- **BM Buthelezi** (KwaBulawayo HS Principal) Mobile: +27725978084
- **Nathi Bhengu** (Electronics Technologist, Tutor, Dataholics leader) Mobile: +27 79 807 9144
- **Bandile Msimango** (Sparks Engineer, Friend) Mobile: 0636179352
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